

Final PatternBoost Experiment Report

Multi-level PatternBoost repository

Final submission run audited July 9, 2026

Abstract

This report records the final audited computational evidence currently stored in the multi-level PatternBoost repository for three geometric extremal-search targets: maximum independent set of rectangles (MISR) LP gaps, unit-square line-stabbing LP gaps, and recursive guillotine hardness. The final submission run used deployed commit `8dd31ca1c8888dff0f1975ccbd38963d73b78b38` on NYUAD Jubail and completed all scheduled Slurm rows without stderr or non-OK Slurm states. The fresh final run certifies score 1.5 for MISR, score 1.5 for unit-square stabbing, and score 0.25 for guillotine hardness. A previously discovered guillotine certificate with score 0.3 is also preserved and reverified under the final deployed code; it is reported separately because the fresh final record array did not rediscover it. All values in the tables below are tied to standalone certificate JSON files and exact verifiers.

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1 Evidence Boundary

The main manuscript track is restricted to:

1. `misr`: LP-gap search for maximum independent set of rectangles;
2. `unit_square`: LP-gap search for stabbing unit squares with horizontal and vertical lines;
3. `guillotine`: hardness for recursive guillotine cutting of disjoint rectangle packings.

The exploratory `epsilon_net` and `graph_separation` tasks are implemented in the repository, but they are appendix material and are not part of the three-problem table in this report. The discarded `square-stabbing-14-9` package is excluded from current evidence.

The final run artifacts are preserved in:

```
docs/assets/final_submission_20260708_131302/
```

This directory contains summaries, audits, best certificate JSON files, renderings, final score tables, and compact learning-curve data.

2 Run Provenance and Audit

Table 1: Final submission run inventory.

Stage	Slurm job	Rows	Status
Smoke	16566068	3	Completed cleanly; used only to release dependencies.
Main matrix	16566072	81	Completed cleanly; 81 summaries, zero stderr.
Controls	16566073	9	Completed cleanly; 9 summaries, zero stderr.
Record follow-up	16566074	12	Completed cleanly; 12 summaries, zero stderr.

The deployed code directory on Jubail was:

```
/home/sg9396/patternboost/multi-level-8dd31ca1c888
```

The run root was:

```
/home/sg9396/patternboost/multi-level-8dd31ca1c888/runs/final_submission_20260708_131302
```

The audit rule is simple: a row is manuscript-usable only when the summary has a certificate path and the certificate recomputes to the summary score under the exact problem-specific verifier. Rows with no best certificate are reported as failed audit rows and are not used as evidence.

Table 2: Certificate audit results.

Stage	Rows	Passed	Failed	Learning rows	Interpretation
Main matrix	81	76	5	490897	The five failures are guillotine rows with no <code>best_certificate_path</code> ; no claimed certificate failed verification.
Controls	9	9	0	53035	All control certificates verified.
Record follow-up	12	11	1	144365	The one failure is the guillotine rect-direct packing/k-subset row with no best certificate.

3 Final Certified Values

Table 3: Fresh final submission results.

Problem	Score	Stage	Configuration	Certificate details
<code>misr</code>	1.5	Main	<code>triangle_free_rect</code> <code>program_coeff_pivot</code>	$n = 12$, $\alpha^* = 6$, $\alpha = 4$.
<code>unit_square</code>	1.5	Main	<code>triangle_free_exact_gap_pressure</code> <code>line_square_incidence</code> <code>primal_dual_lines</code>	23 squares, $\tau = 4$, $\tau^* = 8/3$.
<code>guillotine</code>	0.25	Main	<code>exact_stab_gap_pressure</code> <code>rect_direct_disjoint</code> <code>recursive_gadget_assembly</code> <code>depth_limited_dp</code>	$n = 8$, saved 6, destroyed 2.

4 Reverified Previous-Best Guillotine Certificate

The fresh final arrays did not rediscover the earlier guillotine score 0.3. However, the earlier certificate is preserved and was recomputed under the final deployed code. It verifies with:

$$n = 10, \quad \text{save} = 7, \quad \text{destroyed} = 3, \quad \text{score} = 3/10 = 0.3.$$

The reverified artifact is:

```
docs/assets/final_submission_20260708_131302/reverified_previous_best/guillotine_0p30_
reverification.json
```

For manuscript use, this certificate should be described as a reverified previous-best certificate, not as a discovery made by the fresh final record array.

Table 4: Record follow-up results.

Problem	Score	Configuration	Certificate details
<code>misr</code>	1.5	<code>triangle_free_rect</code> <code>lp_dual_pivot</code> <code>triangle_free_exact_gap_pressure</code>	$n = 12$, $\alpha^* = 6$, $\alpha = 4$.
<code>unit_square</code>	1.5	<code>line_square_incidence</code> <code>coord_mutation</code> <code>exact_stab_gap_pressure</code>	20 squares, $\tau = 4$, $\tau^* = 8/3$.
<code>guillotine</code>	0.25	<code>recursive_obstruction_grammar</code> <code>recursive_gadget_assembly</code> <code>depth_limited_dp</code>	$n = 8$, saved 6, destroyed 2.

Table 5: Controls.

Problem	Score	Configuration	Certificate details
<code>misr</code>	1.3333333333	<code>quadratic_program_rectangles</code> <code>program_coeff_pivot</code> <code>triangle_free_exact_gap_pressure</code>	$n = 8$.
<code>unit_square</code>	1.4285714286	<code>sqstab_exact_grid</code> <code>sqstab_local_hillclimb</code> <code>exact_stab_gap_pressure</code>	16 squares, $\tau = 4$.
<code>guillotine</code>	0.25	<code>recursive_obstruction_grammar</code> <code>witness_breaking</code> <code>k_subset_nonseparability</code>	$n = 8$, saved 6, destroyed 2.

5 Learning Curves

The compact source table for Figure 1 is:

```
docs/assets/final_submission_20260708_131302/plots/learning_curves_by_problem.csv
```

The full event streams remain on HPC and are not committed to the repository.

Best certified score over PatternBoost generations

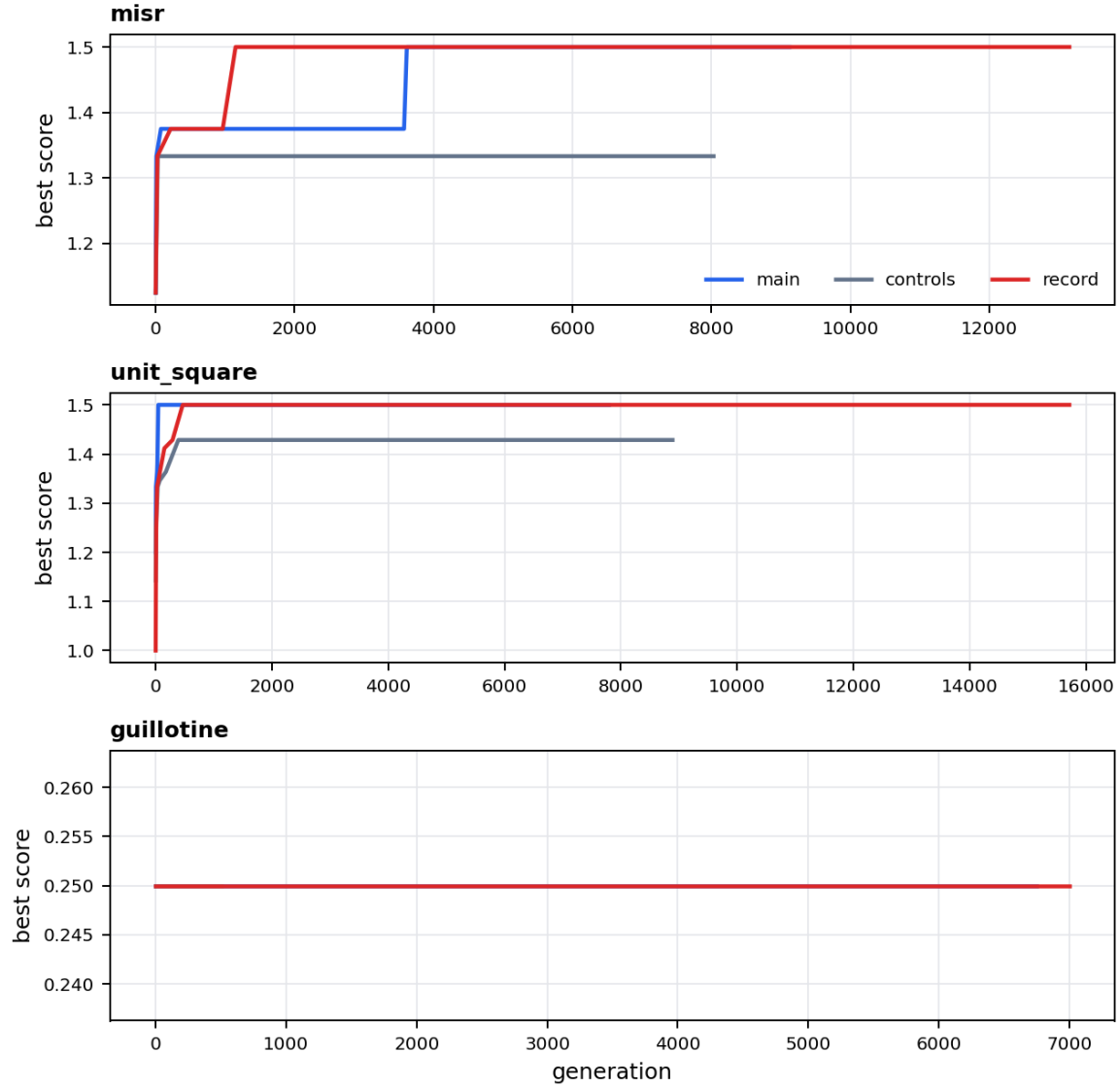


Figure 1: Best certified score over PatternBoost generations for the final submission run. MISR and unit-square stabbing reach score 1.5 in both the main or record stages. Guillotine remains at 0.25 in the fresh final run.

6 Construction Gallery

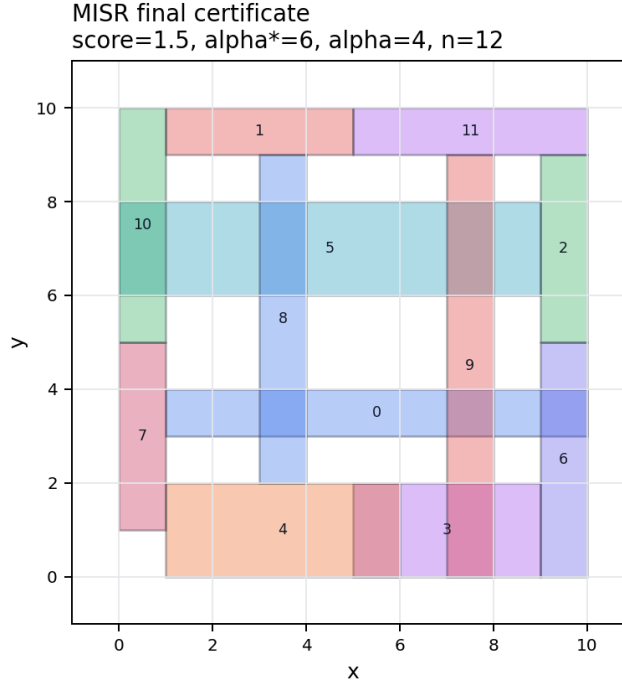


Figure 2: Final MISR certificate from the main matrix. The exact verifier computes $\alpha^* = 6$ and $\alpha = 4$, giving score 1.5.

7 Methodology by Problem

7.1 MISR

For a rectangle family $\mathcal{R}$, the reported score is

$$\text{score}_{\text{MISR}}(\mathcal{R}) = \alpha^*(\mathcal{R})/\alpha(\mathcal{R}),$$

where α^* is the clique-LP optimum and α is the exact maximum independent set. The verifier recomputes the rectangle intersection graph, LP value, integer optimum, maximal clique constraints, and certificate hash.

The final run confirms that the strongest current fresh certificates come from structured sparse-conflict search. The main winning row uses `triangle_free_rect` with `program_coeff_pivot`; the record follow-up ties it with `lp_dual_pivot`. Both certify score 1.5 on 12 rectangles. This is stronger than the current controls, which stop at $4/3$.

7.2 Unit-Square Stabbing

For a unit-square family $\mathcal{S}$, the reported score is

$$\text{score}_{\text{stab}}(\mathcal{S}) = \tau(\mathcal{S})/\tau^*(\mathcal{S}),$$

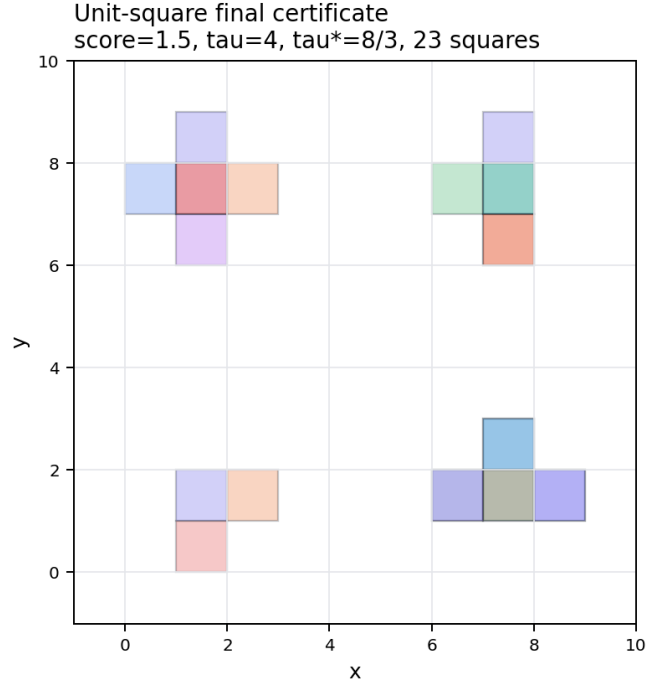


Figure 3: Final unit-square stabbing certificate from the main matrix. The exact set-cover verifier computes $\tau = 4$ and $\tau^* = 8/3$, giving score 1.5.

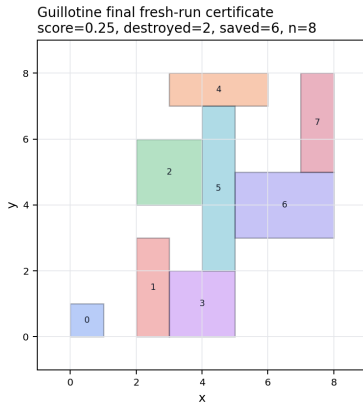


Figure 4: Fresh final guillotine certificate, score 0.25.

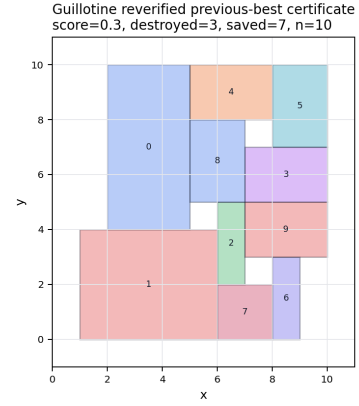


Figure 5: Reverified previous-best guillotine certificate, score 0.3.

where τ is the exact minimum number of horizontal and vertical stabbing lines and τ^* is the finite set-cover LP optimum over critical lines.

The final run certifies score 1.5. The main matrix reaches it with the line-square incidence representation and the primal-dual line move; the record follow-up ties it with line-square incidence plus coordinate mutation. The line-incidence representation is the useful ingredient: it exposes the set-cover structure directly instead of relying only on raw coordinate mutation.

7.3 Guillotine

For a disjoint rectangle packing, the exact verifier computes the maximum number of rectangles that can be saved by a recursive guillotine cutting strategy. The reported score is

$$\text{score}_{\text{guillotine}} = 1 - \text{save}/n = \text{destroyed}/n.$$

The fresh final run certifies score 0.25 in both main and record stages. The stronger previous-best score 0.3 is available only as a reverified historical certificate in this repository snapshot. The final record row that matched the old 0.3 configuration,

`rect_direct_disjoint/packing_resize/k_subset_nonseparability,`

ended with no best certificate under the final run. In the older 36-hour run, the 0.3 certificate appeared late, after about 18 hours. This suggests that the construction family is viable but sparse under fresh stochastic search.

8 Manuscript-Use Recommendations

1. Use the fresh final run as the main reproducible experiment table: MISR 1.5, unit-square 1.5, guillotine 0.25.
2. If the manuscript wants the strongest available guillotine certificate, quote 0.3 only with the explicit provenance that it is a reverified previous-best certificate, not rediscovered by the final fresh array.
3. Do not quote rows that failed audit because they lack `best_certificate_path`.
4. Use Figure 1 for training-progress charts and Figures 2–5 for construction illustrations.
5. Keep exploratory `epsilon_net` and `graph_separation` results in an appendix or separate section.

9 Artifact Index

Table 6: Repository artifacts for the final report.

Artifact	Path
Main summary	docs/assets/final_submission_20260708_131302/main_results/ summary.csv
Main audit	docs/assets/final_submission_20260708_131302/main_results/ audit/audit.json
Controls summary	docs/assets/final_submission_20260708_131302/control_results/ summary.csv
Controls audit	docs/assets/final_submission_20260708_131302/control_results/ audit/audit.json
Record summary	docs/assets/final_submission_20260708_131302/record_results/ summary.csv
Record audit	docs/assets/final_submission_20260708_131302/record_results/ audit/audit.json
Final score table	docs/assets/final_submission_20260708_131302/final_score_ table.csv
Learning curves	docs/assets/final_submission_20260708_131302/plots/learning_ curves_by_problem.csv
Best certificates	docs/assets/final_submission_20260708_131302/certificates/
Best renderings	docs/assets/final_submission_20260708_131302/renderings/
Reverified guillotine 0.3	docs/assets/final_submission_20260708_131302/reverified_ previous_best/